

Values, Points, and Exact Comparisons in Polynomial Optimization

Research report, October 2026

Abstract

Polynomial optimization has several distinct output problems: an objective gap, distance to the optimizer set, approximation of one fixed optimizer, exact comparison, and an exact representation. We develop these distinctions in one bit-complexity framework. For sparse globally convex polynomials over arbitrary rational polyhedra, we give an effective global error bound and a minimum-norm point oracle polynomial in input length, numerical degree, and requested precision. For residual-convex cubics on product boxes, finite noise on the nonconvex core supports full selected-point approximation at every precision without a joint core convexifier. For strongly convex quartics under arbitrary rational linear constraints, exact comparison is ordinary fixed-parameter tractable in the nonlinear dimension of the objective. Structured box and network-flow problems admit deterministic polynomial-time comparison with a PosSLP oracle. An unconstrained exact upper reduction extends to unary degree, and a circuit representation gives short interior Grams despite exponential expanded-output lower bounds. We combine these extensions with audited point-hardness and exact-quartic results, identify their dependencies on prior work, and state the remaining unrestricted constrained comparison problem precisely.

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1 The output is part of the optimization problem

Let f be a rational polynomial and let P be a rational polyhedron. When the minimum is finite and attained, write $f_* = \min_P f$ and $S = \arg \min_{x \in P} f(x)$. This report distinguishes the following tasks.

- (i) A *value-gap point* is a feasible rational point x with $f(x) - f_* \leq \varepsilon$. A certified value interval additionally provides a justified lower bound $\ell \leq f_*$ with $f(x) - \ell \leq \varepsilon$.
- (ii) A *point approximation to the optimizer set* satisfies $x \in P$ and $\text{dist}(x, S) \leq \varepsilon$. Its objective guarantee is a separate statement unless a continuity bound is also provided.
- (iii) A *fixed-selector oracle* approximates the same point $p \in S$ for every requested precision. In the convex setting our selector is the unique point of minimum original Euclidean norm in S .
- (iv) An *exact comparison* decides an order or equality relation, including cases with zero slack. A sufficiently small numerical residual does not decide equality.
- (v) An *exact representation* describes the optimizer or optimum without numerical error. A polynomial system with a unique designated solution can be short even when expanded rational fractions or algebraic coordinates are long.

Our precision parameter is q , with $\varepsilon = 2^{-q}$. Polynomial dependence on q is different from polynomial dependence on $1/\varepsilon$. A fixed-selector oracle is a Cauchy name: the algorithm returns rational approximations at successively requested accuracies, without claiming a short expanded representation of the limiting point.

Input and computation. Rational coefficients and constraint entries have ordinary binary encoding. An explicitly sparse polynomial lists each nonzero coefficient and its exponent vector; repeated monomials are combined. A bound polynomial in the numerical degree D is a bit-polynomial bound when D is unary encoded or polynomially bounded by the remaining input size. It is not a bound polynomial in a binary degree encoding. A succinct arithmetic circuit is a different input format and is not silently substituted for the explicit coefficient list.

An ordinary bit algorithm manipulates explicitly encoded numbers. A shared arithmetic circuit stores the operations defining a number without expanding it. The problem PosSLP asks whether the integer output of a circuit with constants $0, 1$ and operations $+, -, \times$ is positive. A polynomial-time reduction to this problem is not an ordinary polynomial-time algorithm for exact comparison. Nor does PosSLP-hardness assert NP-hardness or that the problem lies outside P.

Structural promises. Convexity on all of $\mathbb{R}^n$, convexity only on P , and a supplied positive strong-convexity modulus are different assumptions. A theorem with a convexity promise does not include a free algorithm for recognizing that promise. A supplied rational matrix identity can sometimes give a polynomially checkable certificate; its encoding and verification cost then belong to the input contract. For nonconvex core models, a supplied joint quadratic convexifier is stronger than convexity of every residual fiber.

Relationship with prior convex polynomial optimization. Slot, Steurer, and Wiedmer prove effective solution bounds and a polynomial-time value-gap algorithm for globally convex polynomials over rational polyhedra. Their structural decomposition uses the polynomial's directions of invariance and linearity. Those tools and the classical convex value-oracle framework are prior work. The point-output questions here ask for effective conversion from objective error to distance, and for a consistent selected point. The precise comparison and the scope of any extension must be made at the theorem level.

How to read this report. The accompanying result map and source audit distinguish historical results from the present extensions. Complete companion proofs specify the reductions and arithmetic constants that are too long to reproduce in every section. Internal independent reviews assess the saved arguments; they are not journal peer review. Exact finite diagnostics check identities and implementation contracts, while the mathematical proofs carry the universal claims.

2 Effective point output for globally convex polynomials

The first positive result converts value accuracy into distance accuracy without a supplied growth modulus. It allows nonunique minima, singular Hessians, unbounded polyhedra, and variable degree. The polynomial must be convex on all of Euclidean space.

Theorem 2.1 (Global point oracle). *Let $f \in \mathbb{Q}[x_1, \dots, x_n]$ be globally convex and explicitly sparse, and let $P = \{x : Bx \leq b\}$ be a rational polyhedron. Write I for the binary input length and $D = \max\{2, \deg f\}$. In $\text{poly}(I, D)$ bit work one can decide emptiness and unboundedness below. Otherwise the minimum is attained, and one can compute positive rationals $R, \Gamma \geq 1$ with*

$$\log R + \log \Gamma = \text{poly}(I, D)$$

such that the minimum-norm optimizer p satisfies $\|p\| \leq R$ and

$$\text{dist}(x, S) \leq \Gamma(f(x) - f_*)^{1/D} \quad (x \in P, 0 \leq f(x) - f_* \leq 1). \quad (1)$$

For every $q \geq 0$, a feasible rational approximation to this same p within 2^{-q} can be computed in $\text{poly}(I, D, q)$ bit work. A simultaneous objective enclosure of width 2^{-s} has work $\text{poly}(I, D, q + s)$.

The numerical degree appears in this bound. With unary degree data it is a polynomial-input-length theorem; with exponentially large binary exponents it is not. Convexity is a promise, or its certification must be charged. The empty and zero-variable cases are direct rational decisions.

Small optimizer norms, attainment, unboundedness detection, and value-gap optimization are already established by Slot et al. [10]. The additional output here uses an effective error bound and a quantitative selector schedule. The radius argument below provides direct rational constants and preserves sparse evaluation. It also makes explicit which prior structural facts are needed. The full constants and representation accounting are in [the companion proof](#).

2.1 A rational quadratic lower bound and a radius

Put $K_D = (D + 1)D^{D+2}$. For $a, x \in \mathbb{R}^n$, the Bregman polynomial

$$h(t) = f(a + t(x - a)) - f(a) - t \nabla f(a)^\top (x - a)$$

is nonnegative, vanishes at zero, and satisfies $0 \leq h(t) \leq h(1)$ on $[0, 1]$. Lagrange interpolation at j/D , $0 \leq j \leq D$, bounds its second derivative at zero by $K_D h(1)$. Hence

$$f(x) \geq f(a) + \nabla f(a)^\top (x - a) + K_D^{-1} (x - a)^\top \nabla^2 f(a) (x - a). \quad (2)$$

Sharper tangent-quadratic bounds are prior; this elementary constant suffices for the encoding estimates.

Average (2) over $a \in [0, 1]^n$. The resulting matrix M , vector ℓ , and scalar c are rational:

$$\begin{aligned} M &= \int \nabla^2 f(a) da, \\ \ell &= \int [\nabla f(a) - 2K_D^{-1} \nabla^2 f(a) a] da, \\ c &= \int [f(a) - \nabla f(a)^\top a + K_D^{-1} a^\top \nabla^2 f(a) a] da. \end{aligned}$$

Each integral is evaluated termwise using $\int a^\alpha da = \prod_i (\alpha_i + 1)^{-1}$. Sparse differentiation and integration have polynomial bit cost in I, D . We obtain

$$f(x) \geq c + \ell^\top x + K_D^{-1} x^\top M x. \quad (3)$$

The PSD matrix M has the same kernel as the common kernel of all Hessians. Indeed, $d^\top M d = 0$ forces the continuous nonnegative polynomial $d^\top \nabla^2 f(a) d$ to vanish on the cube, hence everywhere. PSD then gives $\nabla^2 f(a) d = 0$. Let Π be the rational orthogonal projection onto $\text{range } M$, and put $w = -(I_n - \Pi) \nabla f(0)$. Integration along kernel directions gives

$$f(x) = f(\Pi x) - w^\top x, \quad \ell = \Pi \ell - w. \quad (4)$$

This nonlinear-subspace and affine-direction distinction is part of the prior structure theory; no transformed polynomial is expanded here.

Integer-minor bounds give an explicit $\rho > 0$ with polynomial bit length such that $K_D^{-1} x^\top M x \geq \rho \|\Pi x\|^2$. For example, clear denominators with Q_M , bound the entries of $Q_M M$ by $C_M \geq 1$, and take

$$\rho = [K_D Q_M (nC_M)^{n-1}]^{-1}.$$

The product of the nonzero eigenvalues of an integer PSD matrix is a positive integer, which proves the claimed bound. If $M = 0$, both sides of the quadratic inequality are zero and the same positive formula is harmless.

Find a rational feasible $a \in P$. The rational linear system

$$w = B^\top \lambda + Mz, \quad \lambda \geq 0 \quad (5)$$

decides boundedness below. If it is infeasible, Farkas' lemma supplies $Bd \leq 0$, $Md = 0$, $w^\top d > 0$; the feasible ray $a + td$ has objective decreasing without bound by (4). If it is feasible, put $v = Mz$, $A = \|\Pi\ell - v\|_1$, and $\beta = c - \lambda^\top b$. Equations (3)–(5) give

$$f(x) \geq \beta - A \|\Pi x\| + \rho \|\Pi x\|^2 \quad (x \in P). \quad (6)$$

This is a finite lower bound. On the sublevel $f(x) \leq f(a)$, it bounds $\|\Pi x\|$ by an explicit polynomial-bit number T . It also bounds $w^\top x$ on both sides: the upper bound follows from $w^\top x \leq \lambda^\top b + v^\top \Pi x$, and the lower bound from

$$w^\top x = f(\Pi x) - f(x) \geq f(0) - \|\nabla f(0)\|_1 T - f(a).$$

The one-sided linear constraints alone do not bound $|w^\top x|$.

Stack M and $w^\top$ into a rational matrix J . For every point y in this sublevel, the slice $Z_y = \{x \in P : Jx = Jy\}$ contains an equally good feasible representative of polynomially bounded logarithmic norm. To see this, clear denominators in J and in the constraint rows. If their integer entries are bounded by C , the right-hand-side-uniform Hoffman estimate gives

$$\text{dist}(a, Z_y) \leq (nC)^{n-1} \|J_Z(a - y)\|, \quad (7)$$

where J_Z is the integer-scaled equality matrix. The right-hand side is bounded using T and the two bounds on $w^\top y$.

For completeness, the Hoffman estimate follows by projecting onto the slice, retaining at most n independent equality and active inequality normals, and bounding their least singular value below by $(nC)^{-(n-1)}$. Their Gram determinant is a positive integer. The signs of the inequality multipliers permit the equality residual alone to bound the projection distance for $a \in P$. An irrational equality right-hand side causes no difficulty.

Equality of J -values preserves f by (4). A minimizing sequence therefore has equally good representatives in a fixed closed ball with rational polynomial-bit radius R . Compactness proves attainment. The minimum-norm optimizer has norm at most R . This argument does not presume attainment in order to prove the radius.

2.2 Sparse coefficient rows control distance

Write $f(x) = \sum_\alpha f_\alpha x^\alpha$. For every exponent γ appearing in a partial derivative, form the row

$$N_{\gamma,i} = (\gamma_i + 1) f_{\gamma+e_i}.$$

Missing coefficients are zero. With s original monomials, there are at most ns rows. The identity

$$\nabla f(z)^\top d = \sum_\gamma (Nd)_\gamma z^\gamma$$

shows that $\ker N$ is the global translation-invariance space. Computing this coefficient space is prior structure; the following quantitative bound is what converts it into a point oracle.

Fix the minimum-norm optimizer p , any $x \in P$, and write $d = x - p$, $E = f(x) - f_* \in (0, 1]$. The case $E = 0$ already has $x \in S$. The polynomial

$$h(u) = f(p + u) - f(p) - \nabla f(p)^\top u$$

is globally nonnegative. Convexity and constrained optimality give $0 \leq h(td) \leq E$ for $0 \leq t \leq 1$. Univariate interpolation gives

$$|h(2td)| \leq (D + 1)D^D E(1 + 2|t|)^D.$$

For $c \in [0, 1]^n$, global convexity further gives

$$0 \leq h(c - p + td) \leq \frac{1}{2}h(2(c - p)) + \frac{1}{2}h(2td).$$

The first term has a rational coefficient-sum bound depending on R and f . On $0 \leq t \leq E^{-1/D}$, the entire right-hand side has a bound of polynomial logarithmic size. Differentiating its univariate interpolation formula at zero yields

$$|\nabla f(c)^\top (x - p)| \leq C_* E^{1/D} \quad (c \in [0, 1]^n), \quad (8)$$

with explicitly computable rational C_* and $\log C_* = \text{poly}(I, D)$. Only p needs a radius bound; x can lie anywhere in the original polyhedron. If $E = 0$, the polynomial along d vanishes identically and the midpoint bound makes the second polynomial constant, so (8) holds with zero right-hand side.

Put $r = D - 1$ and $B_r = (r + 1)2^r r^r$. Tensor Lagrange interpolation shows that each coefficient of a polynomial of individual degree at most r is bounded by B_r^n times its supremum on $[0, 1]^n$. This conceptual grid is never enumerated: its only algorithmic consequence is the integer B_r^n , whose bit length is $O(nD \log(D + 1))$. Consequently

$$|(N(x - p))_\gamma| \leq B_r^n C_* E^{1/D}.$$

At zero gap this proves that every optimizer has the same N -value. Conversely, invariance preserves the objective, giving

$$S = \{x \in P : Nx = Np\}. \quad (9)$$

Clear denominators with Q_N , and bound the integer entries of $Q_N N$ and the scaled constraint rows by $C_N \geq 1$. The Hoffman estimate gives (1), for example with

$$\Gamma = \max\{1, (nC_N)^{n-1} Q_N \max(1, \#\text{rows of } N) B_r^n C_*\}.$$

Every operation uses sparse input and rational numbers of polynomial bit length. A constant objective gives an empty coefficient matrix and $S = P$, which is covered without a rank assumption.

2.3 Selecting one fixed optimizer

Let $\varepsilon = 2^{-q}$ and set

$$\tau = \frac{\varepsilon^{2D-2}}{4 \cdot 8^{D-1} R^D \Gamma^D}, \quad \eta = \frac{\tau \varepsilon^2}{4}.$$

The minimizer x_τ of $f(x) + \tau \|x\|^2$ over P exists, is unique, and satisfies $\|x_\tau\| \leq \|p\| \leq R$ by comparison with p . Let s be a nearest optimizer to x_τ , and put $e = \|s - x_\tau\|$. Since $e \leq 2R$, comparison with s and (1) give

$$e^D / \Gamma^D \leq f(x_\tau) - f_* \leq \tau e(2\|x_\tau\| + e) \leq 4R\tau e.$$

Thus $e \leq \varepsilon^2/(8R)$. The projection inequality $p^\top(s - p) \geq 0$ and $\|x_\tau\| \leq \|p\|$ imply

$$\|x_\tau - p\|^2 \leq 2Re \leq \varepsilon^2/4.$$

The coefficient $4R$ allows the nearest optimizer s to lie outside the radius box.

Optimize the regularized polynomial over $Q = P \cap [-R, R]^n$ to certified gap η . Its modulus is 2τ , so the returned feasible rational point is within $\varepsilon/2$ of x_τ , hence within ε of p . The required coefficient and accuracy lengths are polynomial in I, D, q . The convex value routine retains the original sparse polynomial during affine-hull reduction and evaluates the composition by the chain rule, rather than expanding it. Classical rational feasibility repair and value certification then apply.

This completes the proof of Theorem 2.1. The theorem does not decide exact active constraints or exact coordinate signs. Those require a different output guarantee even though arbitrarily accurate rational points are available.

3 Convexity on the feasible domain: a cubic–quartic boundary

Global convexity is a substantive assumption in the preceding point theorem. Convexity only on a polytope still suffices for cubics, with effective constants and a fixed selector. For quartics, even a constant accuracy point query can decide difficult arithmetic comparisons. These statements concern the explicitly encoded polynomial and the output contracts of Section 1; they do not assert that the underlying arithmetic problems are outside polynomial time.

3.1 Effective point approximation for cubics on polytopes

Theorem 3.1 (Domain-convex cubic point oracle). *Let $P = \{x \in \mathbb{R}^n : Cx \leq b\}$ be a nonempty bounded rational polytope. Let $f \in \mathbb{Q}[x]$ be an explicitly supplied polynomial of degree at most three, promised convex on P . If I is the total binary input length, a deterministic algorithm computes a rational $\Gamma_P \geq 1$ with $\text{poly}(I)$ bits such that, for $S = \arg \min_P f$,*

$$\text{dist}(x, S) \leq \Gamma_P (f(x) - f_*)^{1/4} \quad \text{when } x \in P, \quad 0 \leq f(x) - f_* \leq 1. \quad (10)$$

For every $q \geq 0$, it returns a feasible rational point within 2^{-q} of S in time $\text{poly}(I + q)$. It can instead approximate the unique minimum-Euclidean-norm point of S to the same accuracy and in the same time bound. The norm is taken in the original coordinates. No uniqueness, strong convexity, or optimizer-interiority assumption is required. Convexity recognition is outside the theorem.

The proof is included at the level needed to identify the effective constants and the role of degree three. The complete coefficient estimates and weak-optimization interface are in the [cubic polytope companion proof](#).

Affine hull and a rational interior ball. For each row of $Cx \leq b$, rational linear programming determines whether its maximum slack on P is zero. These universally tight rows define $\text{aff}(P)$: averaging a strict-slack witness for every other row gives a point strict in all of those rows at once. Rational elimination therefore gives

$$x = x_0 + Vu, \quad P' = \{u : Au \leq d\}, \quad \phi(u) = f(x_0 + Vu),$$

with P' full dimensional and bounded. Choose free original coordinates as u , so that V contains an identity submatrix. All these data have polynomial binary length. Explicit substitution preserves polynomial size because the degree is at most three. The zero-dimensional case is immediate.

For each nonzero row $a_i^\top$ of A , impose $a_i^\top c + \rho \|a_i\|_1 \leq d_i$, and maximize ρ subject to $0 \leq \rho \leq 1$. Full dimension gives a positive rational optimum. Thus $B(c, \rho) \subseteq P'$. Coordinate linear programs also give a rational $R \geq 1$ with $\|u - c\| \leq R$ on P' . The logarithms of R and $1/\rho$ are polynomially bounded. This reduction must precede the Hessian argument: convexity on a lower-dimensional set need not imply positive semidefiniteness of the ambient Hessian.

An affine Hessian identifies the relevant directions. Write $H(u) = \nabla^2 \phi(u)$ and $H_c = H(c)$. The matrix $H(u)$ is affine in u and positive semidefinite on P' . For each $u \in P'$, the reflected point $u' = c - (\rho/R)(u - c)$ lies in the interior ball. Taking the affine combination that equals c gives

$$0 \preceq H(u) \preceq \beta H_c, \quad \beta = 1 + R/\rho. \quad (11)$$

Consequently $K = \ker H_c = \bigcap_{u \in P'} \ker H(u)$ is a known rational subspace. Boundary Hessians may have larger kernels. Choose a rational $M \geq 1$ bounding $\|H(u)\|$ on P' ; for example, $M = \max\{1, \beta \max_i \sum_j |(H_c)_{ij}|\}$ works. If $H_c = 0$, the objective is affine and linear programming handles optimization; the same linear error-bound argument below supplies (10) in this branch.

Let v be any optimizer of ϕ , $u \in P'$, $d = u - v$, and $E = \phi(u) - \phi(v)$. Put $a = d^\top H(v)d$ and $b = d^\top H(u)d$. Exact cubic Taylor expansion and first-order optimality imply

$$E = \nabla \phi(v)^\top d + (2a + b)/6 \geq (a + b)/6.$$

Full symmetry of the constant third derivative gives the additional identity

$$d^\top H_c d = a + (c - v)^\top (H(u) - H(v))d.$$

For a positive semidefinite matrix Q , $\|Qd\| \leq \sqrt{\|Q\| d^\top Q d}$. Combining this inequality with the diameter bound $2R$ gives the conservative estimate

$$E \geq \frac{(d^\top H_c d)^2}{384MR^2} \geq \frac{\lambda^2 \|\Pi d\|^4}{384MR^2}, \quad (12)$$

where Π projects onto $K^\perp$ and λ is the smallest positive eigenvalue of H_c . The unknown optimizer and this projector are used only in the analysis.

The optimizer set is an affine slice. Put $g_c = \nabla \phi(c)$. Integrating the Hessian shows that $\nabla(\phi - g_c^\top u)$ belongs to $K^\perp$ throughout P' . Equation (12) then proves

$$\arg \min_{P'} \phi = \{w \in P' : H_c(w - v) = 0, g_c^\top(w - v) = 0\}. \quad (13)$$

The gradient row is needed because ϕ can have a nonzero affine slope along the common Hessian kernel. It satisfies

$$|g_c^\top d| \leq E + MR \|\Pi d\|.$$

The slice right-hand side may be irrational; its coefficient matrix is known and rational.

Here is the quantitative linear error bound used for that slice. If $Q = \{u : Bu \leq h\}$ and $Z = \{u \in Q : Tu = t\} \neq \emptyset$, with integer matrices B, T having s columns and entries bounded in absolute value by $C_* \geq 1$, then

$$\text{dist}(u, Z) \leq (sC_*)^{s-1} \|Tu - t\| \quad (u \in Q). \quad (14)$$

Neither right-hand side has to be rational. To see the stated height bound, project u onto Z and express the displacement using independent equality normals and a nonnegative combination of active inequality normals. Eliminate dependence modulo the equality span. The resulting integer row matrix has at most s independent rows; its Gram determinant is at least one and its singular values are at most sC_* . Its smallest singular value is therefore at least $(sC_*)^{-(s-1)}$. Taking the inner product with the displacement discards the nonpositive contributions of active outward inequality normals, since $u \in Q$, and proves (14).

Clear denominators of H_c, g_c, A by a positive integer Δ . If H_c has positive rank, a nonzero principal minor gives the rational eigenvalue lower bound $\lambda \geq \Delta^{-s} M^{-(s-1)}$. Equations (12)–(14) now give a computable constant of polynomial binary length multiplying $E^{1/4}$ for $E \leq 1$. Multiplication by a rational upper bound on $\|V\|$ proves (10) in original coordinates. When $H_c = 0$, the gradient row alone gives an error bound proportional to E , hence also to $E^{1/4}$ on this range.

Feasible output and a consistent selector. Classical weak convex optimization applies to the capped epigraph with the known inner and outer radii [6]. Approximate feasibility must be repaired. If a rational u is within δ of P' , the rational point

$$\hat{u} = \frac{\rho u + \delta c}{\rho + \delta}$$

lies in P' exactly. Relative to a nearby feasible point the repair costs at most $\delta(1 + R/\rho)$. A coefficient bound on $\|\nabla\phi\|$ controls its objective cost. Together with the usual erosion bound for the capped epigraph, this yields certified value intervals of width η in time $\text{poly}(I + \log(1/\eta))$. Choosing $\eta = (2^{-q}/\Gamma_P)^4$ proves the set-distance statement.

For completeness, let p be the minimum-original-norm optimizer and let $R_x \geq 1$ bound $\|x\|$ on P . Set

$$\varepsilon = 2^{-q}, \quad \tau = \frac{\varepsilon^6}{1024R_x^4\Gamma_P^4}, \quad \eta = \tau\varepsilon^2/4.$$

Minimize $f(x) + \tau\|x\|^2$ to feasible value gap η . If x_τ is its exact minimizer and $e = \text{dist}(x_\tau, S)$, projection onto the closed convex set S and regularized optimality give

$$e^4/\Gamma_P^4 \leq f(x_\tau) - f_* \leq 2R_x\tau e, \quad \|x_\tau - p\|^2 \leq 2R_x e.$$

The gap is at most $\tau R_x^2 \leq 1$, so the error bound is applicable. These inequalities give selection bias at most $\varepsilon/2$. Strong convexity of the regularized objective gives another $\varepsilon/2$ for its value approximation. In reduced coordinates the penalty is $\tau\|x_0 + Vu\|^2$; replacing it with $\tau\|u\|^2$ could select a different optimizer.

The exponent $1/4$ is sufficient for polynomial dependence on precision bits. The example $f(t) = t^3$ on $[0, 1]$ excludes a uniform error exponent larger than $1/3$. The present argument does not prove that $1/3$ is attainable with effective constants in all dimensions. That sharper exponent is not needed for Theorem 3.1.

3.2 Quartics can encode an arithmetic sign in one endpoint

Theorem 3.2 (Constant-accuracy point reductions). *There are polynomial-time reductions with one point-oracle query from each of the following decision problems to finding a point within Euclidean distance $1/4$ of the optimizer of an explicitly encoded rational quartic convex on a rational box:*

- (i) *Square Root Sum, with positive integer radicands and an integer threshold. The output instance has interaction-treewidth at most two, bounded coefficient magnitudes, and bounded upper coordinate curvature.*

(ii) PosSLP. *The output instance has polynomial encoding length and can have bounded coefficient magnitudes; no bounded-treewidth claim is made.*

Both instances can be written on unit boxes and have unique optimizers. In both, a designated optimizer coordinate equals zero or one according to the source comparison. Neither reduction supplies a uniform effective lower growth bound for the final objective.

The complete constructions are the [radical point reduction](#) and the [arithmetic-circuit point reduction](#). The following common step explains why the accuracy remains constant even when the arithmetic sign has extremely small magnitude.

Suppose two independent strongly convex base problems contain coordinates $a_+, a_- \in [0, 1]$, with exactly one positive at their respective unique minima. The positive coordinate records which side of a strict comparison holds. With a fresh $y \in [0, 1]$, add

$$\eta[a_+^2(y-1)^2 + a_-^2y^2], \quad \eta = 1/192. \quad (15)$$

For a direction (p, q) in (a, y) , direct differentiation gives

$$D^2[a^2(y-c)^2][(p, q)]^2 = 2(aq + 2(y-c)p)^2 - 6(y-c)^2p^2.$$

The possible negative contributions are absorbed by the base curvature, which is at least $1/16$ in the radical construction and at least $1/2$ in the unscaled circuit construction. This proves joint convexity on the box, including where an amplitude vanishes.

Every full objective is bounded below by the sum of its two base minima. If $a_+^* > 0$, choosing both base minimizers and $y = 1$ attains this bound; if $a_-^* > 0$, choose $y = 0$. Every minimizer must therefore attain both unique base minima and make (15) zero. The full optimizer is unique and its endpoint is forced. Thresholding an approximation to y at $1/2$ decides the source comparison. Feasibility of that approximation is not needed for the implication.

In the radical reduction, normalized cubic leaf objectives have minimizers $\sqrt{a_i}/R_i \in [1, 2]$, and a binary averaging tree propagates their weighted sum. Opposite linear couplings to the root produce the two amplitude tests. Equality is removed by polynomial-time integer square-root tests: a positive sum of square roots of integers can be an integer only when all its summands are integers. One proof takes traces in the common multiquadratic extension; every nonsquare radical has trace zero, and positivity excludes their remaining sum being zero. The algorithm does not construct that extension or factor the radicands. The tree factors need bags of size at most three, and the two amplitude-endpoint edges join the two trees without increasing that bound. After rescaling to the unit box, remove the aggregate constant term; this preserves the minimizer and the bounded-coefficient promise.

In the circuit reduction, bounded rational-pair gate values encode the integer circuit without expanding its potentially enormous output. Triangular degree-two residual equations have a unique common zero. Geometrically decreasing weights on their squares dominate the negative Hessian terms and produce a quartic strongly convex on its box. The final signal has the sign of $2A - 1$, where A is the integer source output, so it never vanishes. Opposite amplitude tests and (15) finish the reduction. The companion proof checks arbitrary fanout and repeated operands. Scaling the entire objective makes its collected coefficient magnitudes bounded while preserving its minimizer; this also scales its curvature.

Before optional whole-objective scaling, the endpoint curvature is $2\eta((a_+^*)^2 + (a_-^*)^2)$ and can be extremely small. Thus a value-gap guarantee does not give the requested point guarantee. Global convexity is also absent: box convexity is exactly what the Hessian estimates establish. These qualifications explain compatibility with the positive global-convex and cubic-domain results.

An ordinary polynomial-time point algorithm for either target class would put its source problem in P. An always-correct algorithm with expected polynomial work would give the corresponding Las Vegas algorithm. These are conditional implications, not NP-hardness claims or proved class separations. The distinction between approximate values and strong approximation of solution coordinates has established precedents in algebraic fixed-point complexity [4]; the specialized convex-polynomial realizations require the separate proofs cited above. Their independent [point-reduction audit](#) and the [positive-core audit](#) record the checked assumptions. Neither audit establishes publication priority.

4 Exact comparisons and succinct arithmetic

Polynomial-time approximation at every requested accuracy does not settle an exact order test. A nonzero quantity may require exponentially many printed accuracy bits to distinguish from zero. For globally strongly convex polynomial optimization, shared Newton circuits reach that accuracy with polynomially many arithmetic operations. Algebraic separation then converts exact comparisons, including equality, into one PosSLP instance.

4.1 One sign circuit for a polynomial observable

Theorem 4.1 (Strong-convexity upper reduction). *Let $n \geq 1$ and let $f, h \in \mathbb{Q}[X_1, \dots, X_n]$ be explicitly supplied sparse polynomials, with unary degree bounds. Suppose the input supplies $\mu \in \mathbb{Q}$, $\mu > 0$, and promises*

$$\nabla^2 f(X) \succeq \mu I \quad (X \in \mathbb{R}^n).$$

Let p be the unique unconstrained minimizer of f . Each strict order, weak order, and equality predicate on $h(p)$ versus zero has a deterministic polynomial-time many-one reduction to one PosSLP instance. The circuit is constructed without PosSLP queries. A rational threshold is absorbed in h and counted in the input. The same conclusion holds under any degree encoding guaranteeing numerical degrees polynomially bounded by the input length.

The degree-four theorem and its observable extension are proved in the [quartic upper-bound companion](#). The explicit sparse, unary-degree extension is proved in the [variable-degree companion](#). The argument below states the common mechanism and its representation requirements.

A polynomial-bit starting point. Let $L \geq 2$ include the dimension, both polynomials, the degree bounds, and μ . Increasing it by a fixed constant if necessary, assume $n, \deg f, \deg h \leq L$, coefficient one-norms at most 2^{2L} , and $2^{-L} \leq \mu \leq 2^L$. Strong convexity gives coercivity and

$$\|p\| \leq \|\nabla f(0)\|/\mu \leq 2^{3L}.$$

Set $R = 2^{4L}$ and $B = 2^{16L^2}$. On $\|X\| \leq R$, B bounds values and derivative operator norms through order three for both polynomials. Indeed, an order- j component is bounded by $2^{2L} L^j R^L$, and passage to a tensor norm costs at most $L^{j/2}$. The third derivative bound makes B a Hessian Lipschitz constant.

Put

$$\rho = \mu/(4B), \quad \varepsilon_0 = \mu^3/(32B^2).$$

These rationals have polynomial binary length. Convex value optimization gives a rational x_0 with $f(x_0) - f(p) \leq \varepsilon_0$ in polynomial bit time; Corollary 1.2 of Slot et al. [10] applies directly. Strong convexity implies $\|x_0 - p\| \leq \rho$. The radius- ρ ball about p is contained in the radius- R ball used

above. This preliminary computation asks for polynomially many explicit accuracy bits and does not use a sign oracle.

A separation bound used only in the proof. For $\alpha = h(p)$, the real solution set of

$$\exists X : \quad \nabla f(X) = 0, \quad z = h(X) \quad (16)$$

is the singleton $\{\alpha\}$. Effective one-block real quantifier elimination supplies a quantifier-free description whose polynomial degrees and coefficient bit lengths are at most $2^{a(L)}$, for a fixed effective polynomial a . This uses the degree estimate $D^{O(n)}$ and $n \log D = \text{poly}(L)$; the primary theorem is Basu [2, Theorem 2.27]. The companion proof specifies its coefficient-height use. The reduction does not execute quantifier elimination.

Some nonzero integer polynomial in that description must vanish at α ; otherwise every sign condition would be locally constant near α , contradicting the singleton. Remove a power of z from that polynomial. Its nonzero integer constant coefficient and the elementary root bound give

$$\alpha \neq 0 \implies |\alpha| \geq g := 2^{-2^{a(L)+1}}. \quad (17)$$

We enlarge a so that $a(L) \geq 32L^2 + 10$. The number g has a short circuit: start with $1/2$ and square $a(L) + 1$ times. No assumption on the dimension of the complex critical locus is needed; (16) isolates one real value.

Newton iteration with shared arithmetic. Take exact rational Newton steps

$$x_{t+1} = x_t - \nabla^2 f(x_t)^{-1} \nabla f(x_t).$$

Within the established neighborhood, $e_t = \|x_t - p\|$ obeys

$$e_{t+1} \leq \frac{B}{2\mu} e_t^2, \quad e_t \leq \frac{2\mu}{B} 2^{-3 \cdot 2^t}.$$

These estimates also keep every iterate in that neighborhood. A positive definite Hessian is solved by rational $LDL^\top$ elimination without pivoting, square roots, or sign queries. Sparse differentiation and monomial evaluation add polynomially many circuit gates per step. Shared nodes preserve the values of previous steps without expanding their rational fractions.

After $k = a(L) + 2$ steps, the bound on ∇h gives

$$|h(x_k) - \alpha| \leq B e_k \leq g/8.$$

Let $\hat{\alpha} = h(x_k)$. Each row of the following table gives a rational circuit positive exactly for its listed predicate:

Circuit output	Predicate
$\hat{\alpha} - g/2$	$\alpha > 0$
$\hat{\alpha} + g/2$	$\alpha \geq 0$
$-\hat{\alpha} - g/2$	$\alpha < 0$
$-\hat{\alpha} + g/2$	$\alpha \leq 0$
$g^2/4 - \hat{\alpha}^2$	$\alpha = 0$

For zero, the approximation is within $g/8$; for a nonzero value, its magnitude is at least $7g/8$. This verifies all equality cases without an unknown nonzero-gap promise.

Represent every rational gate as N/D with $D > 0$. In particular, division can be removed by

$$\frac{N_a/D_a}{N_b/D_b} = \frac{N_a D_b N_b}{D_a N_b^2}.$$

All divisors are nonzero, by the positive-definite elimination argument. Ordinary fraction rules handle the other operations, and each rational gate adds only constantly many integer gates. Building printed binary constants from 0 and 1 leaves a polynomial-size integer circuit. Its final numerator is the required single PosSLP instance. This proves Theorem 4.1.

The sparse input and degree bound are essential to the warm start and the ordinary bit cost. This proof does not give a polynomial bound in sparse binary-degree length alone or for a succinct circuit encoding of the input polynomial. It also does not remove the supplied global strong-convexity modulus.

4.2 Checkable curvature and matching quartic lower bounds

A useful input class replaces the semantic curvature promise with a supplied rational identity

$$v^\top \nabla^2 f(X) v = w(X, v)^\top M w(X, v), \quad M \succ 0, \quad (18)$$

where the monomial vector w is linear in v and contains all v_i . The matrix and monomial list are included in the input. Sparse coefficient comparison checks the identity, and rational symmetric elimination checks positive definiteness in polynomial time. If $m = \dim M$, then

$$\mu = \det(M) / (\text{tr } M)^{m-1} > 0$$

is a rational lower bound on $\lambda_{\min}(M)$ of polynomial binary length. Because $\|w(X, v)\|^2 \geq \|v\|^2$, it also bounds the Hessian below. Enlarge L to include this derived μ ; the enlarged length is polynomial in the certificate input length. For quartics we use the full basis $w = (v, X \otimes v)$. Invalid certificates can be rejected before applying the reduction, giving an ordinary decision language. The construction does not recognize all strongly convex polynomials.

Theorem 4.2 (Certified quartic order comparisons). *For rational quartics on $\mathbb{R}^n$ supplied with a positive definite rational Hessian Gram on $(v, X \otimes v)$, exact strict and weak minimum-value comparisons and minimizer-coordinate comparisons are PosSLP-complete under polynomial-time many-one reductions. The minimum-sign lower reduction has nonzero minimum on both source branches. Equality has the upper bound of Theorem 4.1; no matching equality lower bound is asserted here.*

The upper bound follows by choosing $h = f - r$ or $h = X_j - r$. The lower reductions require a globally convex realization of arithmetic computation, which is a separate substantive construction. The [signed-root sign compiler](#) simulates a source integer circuit with bounded analytic signals, and the [quartic realization](#) places those signals at a unique zero of a rational sum of squares. A carefully weighted exposing quadratic controls the otherwise negative Hessian terms of squared residuals. The construction supplies a positive definite Gram on the full Hessian basis, not merely a positive semidefinite form evaluated on rank-one tensor vectors.

For minimum-sign comparison, a cubic perturbation encodes the signal in the minimum without losing the full Hessian certificate. The [unconstrained sign reduction](#) uses distinct signal coordinates

u, v , with $u > 0$, $v \neq 0$, $|v| \leq 1/8$, and $u^2 \leq |v|/2$ at the old zero. After scaling, the perturbed objective has Hessian at least I . At that zero its value is $-u^2v$ and its squared gradient norm is $4u^2v^2 + u^4$. If $v > 0$, the value is already negative. If $v < 0$, strong convexity bounds the possible further decrease by half the squared gradient norm, leaving a strictly positive minimum. The scaling, the signal generator, and the full rational Gram all have polynomial encoding length; they do not require printing the tiny signals. The complete construction and its [independent lower-bound review](#) provide these quantitative interfaces.

The architecture of using accurate Newton circuits, algebraic separation, and division elimination is established prior work [1, 5]. Exact semidefinite feasibility already has arithmetic-circuit comparison reductions [11]. The content of Theorem 4.2 is the classification for the stated explicit quartic class and its strong certificates; it is not a claim that exact convex feasibility or compressed Newton simulation was first studied here.

4.3 A rational optimizer and a known minimum do not settle coordinates

Theorem 4.3 (Rational-optimizer restriction). *The coordinate-comparison lower bound in Theorem 4.2 holds even with all of the following additional promises:*

- (i) *the unique optimizer p lies in $\mathbb{Q}^N \cap [-1, 1]^N$;*
- (ii) *the minimum is the known value zero, and the input supplies $N + 1$ rational quadratic square factors;*
- (iii) *$\nabla^2 F \succeq (3/2)I$, with a supplied full positive definite rational Hessian Gram;*
- (iv) *the queried coordinate is nonzero and the threshold is zero.*

Rationality is a promise, not a property recognized by the theorem.

The [complete reduction](#) first compiles an integer source circuit with output A into a shared circuit of rational unit quaternions. Multiplication and inversion preserve rationality and bounded coordinates. A designated coordinate is nonzero with the sign of $2A - 1$. A specialized quartic realization has as its unique zero exactly the list of quaternion gate values, with their coordinates preserved. It supplies the stated square factors and full Hessian Gram in polynomial time. The [independent review](#) checks the sign compiler, realization, rational certificate, and encoding bounds.

Every optimizer coordinate is rational, but expanding the gate fractions is not part of the construction. Thus the result distinguishes field membership from representation size and arithmetic comparison cost. It concerns an optimizer coordinate. The minimum is already known to be zero, and the theorem does not claim hardness of its sign under this same promise. A perturbation encoding a minimum sign can move the optimizer to an irrational point.

For the general upper theorem, the Newton vector x_k is likewise a rational vector represented by a polynomial-size shared circuit. When $\min f < 0$, choosing $h = f$ ensures $f(x_k) < 0$, so the construction supplies a short rational-circuit feasible witness without first deciding feasibility. Checking its exact objective sign is still the PosSLP comparison. This supplies neither polynomial-size expanded fractions nor a rational feasible point when the minimum equals zero.

4.4 Exact witnesses: existence and expanded size

Even a strongly convex quartic with rational minimum need not have a rational optimizer. The following small instance makes the distinction between a point approximation and an exact feasible witness explicit.

Proposition 4.4 (An irrational zero in two variables). *Put*

$$\begin{aligned} A(x, y) &= 12599x^2 - 10000xy + 7937y^2 - 15874x - 12599y + 20000, \\ F(x, y) &= A(x, y)^2 + 10000((x^2 - y)^2 + (y^2 - 2x)^2). \end{aligned}$$

Then F is a rational sum of squares with a supplied positive definite rational full Hessian Gram, and

$$\nabla^2 F \succeq 4096I, \quad \min_{\mathbb{R}^2} F = 0, \quad \arg \min_{\mathbb{R}^2} F = \{(\sqrt[3]{2}, \sqrt[3]{4})\}.$$

Consequently $\{F \leq 0\}$ is nonempty and contains no rational point. The same is true when the domain is $[1, 2]^2$.

Any zero must satisfy $y = x^2$ and $y^2 = 2x$, hence $x(x^3 - 2) = 0$. The origin is excluded by $A(0, 0) = 20000$. At (r, r^2) with $r^3 = 2$, substitution gives $A = 0$. This proves the zero-set assertion. The curvature statement is an exact rational identity, provided in the [full Hessian certificate](#): after an invertible rational affine change, the Hessian biform minus $4096\|v\|^2$ is represented by a strictly positive definite rational matrix on all six monomials $(v_1, v_2, xv_1, xv_2, yv_1, yv_2)$. Its displayed diagonal-dominance certificate and invertible congruence prove positivity; numerical matrix tests are not part of the argument. The [independent singleton audit](#) checks this identity and the zero-set proof.

The example answers negatively the rational-feasible-point witness question for quartics in Table 1 of Slot et al. [10], version 1. It is consistent with that paper’s univariate result: the present quartic is bivariate. Its optimizer has a short algebraic description, so the absence of a rational point excludes neither other certificate formats nor NP membership of the decision problem. A full positive definite *ordinary polynomial* Gram cannot represent this F , because it vanishes at a real point and the ordinary monomial vector contains the constant one. Its positive definite *Hessian* Gram represents a different polynomial in a different basis.

Theorem 4.5 (Expanded witness bounds). *The following assertions concern ordinary binary fractions.*

- (i) *There are polynomial-size rational quartics in $n = 2k$ variables, with supplied full Hessian Grams at least the identity, whose zero sublevel sets are bounded and have nonempty interior, but every rational feasible point requires $\Omega(n2^{n/2})$ bits. All feasible coordinate magnitudes are bounded by an absolute constant.*
- (ii) *There are strictly positive rational quartics with such Hessian certificates that admit both a short rational positive semidefinite ordinary Gram and a rational positive definite ordinary Gram. Every rational positive definite ordinary Gram requires $\Omega(n2^{n/2})$ total denominator bits. The basis is the full vector of monomials of degree at most two, each occurring once.*
- (iii) *Every strictly positive rational quartic supplied with a positive definite full rational Hessian Gram has a positive definite rational ordinary Gram with total expanded size $\text{poly}(L)2^{O(n)}$, where L counts the polynomial and supplied Hessian Gram. This is a size bound, not a polynomial-time construction in L alone.*

The complete proofs are the [feasible-point lower bound](#), [positive definite Gram lower bound](#), and [Gram size upper bound](#). The lower constructions encode a positive signal at most $1000^{-(k+3)2^k}$ using k bounded cubic-root gates. One perturbation confines the whole feasible body extremely close to a cubic irrational coordinate; the integer difference of cubes then forces a large rational

denominator. Another perturbation produces a strictly positive polynomial with an extremely small value. A uniform trace bound for its ordinary Grams and that small Rayleigh quotient force a tiny positive determinant, hence long denominator encodings. This second argument requires positive definiteness; the same family has a short singular Gram.

For the upper bound, rational sampling in a nonempty open semialgebraic set supplies a center of singly exponential size where $f - \|\nabla f\|^2/(2\mu) > 0$. Taylor integration of the supplied Hessian Gram then gives an ordinary positive definite rational Gram with the claimed size. The [witness audit](#) checks the sampling theorem, the full-span Taylor construction, and both lower bounds. Exponential dependence on dimension is therefore qualitatively necessary and sufficient for these interior Grams; the exponent constants are not matched. The lower bounds are superpolynomial in the constructed input length, but are not asserted as $2^{\Omega(L)}$. None excludes a compact arithmetic or algebraic description. Section 7 states the corresponding certificate interfaces.

Finally, a globally convex polynomial of degree at most three is quadratic: its affine Hessian must remain positive semidefinite in both directions of every line, forcing each nonconstant matrix coefficient to vanish. Strongly convex rational quadratic minimization on all of $\mathbb{R}^n$ reduces to rational linear algebra. Degree four is therefore the first degree supporting the constructions above. This degree statement and PosSLP-completeness do not prove a separation from P, an NP-hardness theorem, or a practical exact solver. The [primary-source comparison](#) records the established methodological precedents and the remaining publication-priority limits.

5 Exact comparison with linear constraints

Arbitrary linear constraints leave a gap in the exact classification. For a globally strongly convex rational quartic over a rational polyhedron, the current general bound in this research program is $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$: the full active set is an unambiguous certificate, but finding it deterministically is unresolved. The [active-set proof](#) checks the gradient inequality without guessing a multiplier support. This section gives three positive extensions with different structural assumptions. They do not establish deterministic P^{PosSLP} for arbitrary constrained quartics.

5.1 Few nonlinear directions, arbitrary constraints

Write $f = q + r$, where q has degree at most two and r consists of the cubic and quartic monomials. Define

$$K_f = \{v : D_v r \equiv 0\}, \quad k = n - \dim K_f. \quad (19)$$

We call k the nonlinear dimension. It is the rank of the coefficient matrix of ∇r : each row collects the coefficients of one monomial across the gradient components. Rational linear algebra therefore computes k and a full-row-rank matrix $U \in \mathbb{Q}^{k \times n}$ such that $r(x) = g(Ux)$. A rational right inverse V gives $g(t) = r(Vt)$. These operations have polynomial bit cost for quartics. This is the essential-variable construction, applied to the cubic and quartic pieces; its homogeneous antecedent is Carlini [3, Proposition 1].

Theorem 5.1 (Exact comparison in nonlinear dimension). *Let $P = \{x : Ax \leq b\}$ be an explicit rational polyhedron, and let $f, h \in \mathbb{Q}[x]$ have degree at most four. Supply a rational $\mu > 0$ with the promise $\nabla^2 f(x) \succeq \mu I$ for every $x \in \mathbb{R}^n$. If L is the total input length and k is (19), a deterministic ordinary bit algorithm runs in $F(k)L^C$, for a computable F and an absolute C , and either reports P empty or decides the sign of $h(p)$ at its unique minimizer p . No exact-comparison oracle is used. The algorithm also recovers the full active set and describes p as a rational affine image of the unique stationary point of a strongly convex rational quartic in at most k variables.*

The constraints may have full rank, redundant rows, empty interior, or unbounded feasible sets. The curvature assumption is a promise; a rational positive semidefinite Gram for $v^\top(\nabla^2 f(x) - \mu I)v$ supplies a checkable subclass. This does not claim that every strongly convex quartic has such a Gram. The exact optimizer description is implicit and need not have rational coordinates. The proof and coefficient bookkeeping appear in the [nonlinear-dimension companion](#).

Elimination on the unknown optimal face. Let S be all tight rows at p . Polyhedral first-order optimality implies that p uniquely minimizes f on the affine space $E = \{x : A_S x = b_S\}$. Choose a rational chart $x = \bar{x} + Zy$. Splitting y along the kernel of UZ gives coordinates (t, w) with $s = \dim t = \text{rank}(UZ) \leq k$, in which

$$f(\bar{x} + ZTt + ZWw) = \frac{1}{2}w^\top Hw + w^\top(Jt + a) + \psi(t). \quad (20)$$

The nonlinear part is independent of w , and $H \succeq \mu(ZW)^\top(ZW) \succ 0$ when this block is nonempty. Thus $w = -H^{-1}(Jt + a)$ and

$$p = u + Dt_*, \quad \phi(t) = f(u + Dt), \quad \eta(t) = h(u + Dt),$$

where D has full column rank and t_* uniquely solves $\nabla\phi(t) = 0$. The reduced objective remains globally strongly convex. All coefficients have uniformly polynomial bit length in L : only a fixed number of rational linear-algebra and degree-four substitution steps are used, on an independent subset of the original rows. The unknown face establishes this bound; it is not guessed or enumerated.

Parameter-dependent separation, then ordinary approximation. The formula

$$\exists t \in \mathbb{R}^s : \quad \nabla\phi(t) = 0, \quad z = \eta(t)$$

defines the singleton $\{h(p)\}$. The coefficient clause of Basu [2, Theorem 2.27] bounds output coefficient bits linearly in the input coefficient-bit bound, multiplied by a function of s for fixed degree. Consequently, for effective G and an absolute polynomial T ,

$$h(p) \neq 0 \implies |h(p)| \geq \gamma := 2^{-G(k)T(L)}. \quad (21)$$

Indeed, a nonzero integer polynomial must vanish at a singleton defined by sign conditions. Removing a power of its variable and comparing its nonzero constant term with the other coefficients gives the root bound. This reasoning neither executes elimination nor assumes a finite complex critical locus. No factor $L^{O(k)}$ enters the coefficient bound.

A polynomial-bit rational feasible point and strong convexity give a polynomial-bit radius containing p . Let $B \geq 1$ bound $\|\nabla h\|$ on its unit enlargement. Set

$$\delta = \min\{1, \gamma/(8B)\}, \quad \varepsilon = \min\{1, \mu\delta^2/2\}.$$

Corollary 1.2 of Slot et al. [10] produces a rational $\hat{x} \in P$ with objective error at most ε in time polynomial in $L + \log(1/\varepsilon)$. Constrained strong convexity gives $\|\hat{x} - p\| \leq \delta$, hence $|h(\hat{x}) - h(p)| \leq \gamma/8$. Comparing the rational value with $\pm\gamma/2$ decides its exact sign, including zero, in ordinary $F(k)L^C$ bit time. Applying this procedure to every slack recovers S ; then (20) constructs the implicit optimizer.

For example, $f(x) = \frac{1}{2}x^\top Qx + c^\top x + (a^\top x)^4$ with $Q \succeq I$ has $k \leq 1$, while its quadratic term and constraint matrix may have full rank. This differs from bounding constraint rank or the rank of the whole objective.

The [existing mixed-linear candidate-list theorem](#) also gives a consequence under the same global joint strong-convexity promise: with t integer variables, exact selection is ordinary FPT in $k + t$. The list contains every optimal integer block in parameter-polynomial size. Fixing such a block leaves nonlinear dimension at most k . Two fiber minima are compared on the product polyhedron using the sum objective and difference observable, whose nonlinear dimension is at most $2k$. This uses the candidate-list theorem as an imported integer search result; it does not supply a new lattice algorithm.

5.2 A transfer principle from exact quadratic subproblems

A second route retains unrestricted nonlinear dimension but requires a suitable arithmetic algorithm for every Taylor quadratic. Let P be a nonempty bounded rational polyhedron with a supplied rational bounding box. For $x \in P$, define

$$N(x) = \arg \min_{y \in P} \left\{ \nabla f(x)^\top (y - x) + \frac{1}{2} (y - x)^\top \nabla^2 f(x) (y - x) \right\}. \quad (22)$$

Theorem 5.2 (Exact-QP arithmetic transfer). *Suppose f is an explicit globally μ -strongly convex polynomial of fixed degree. Assume (22) can be solved exactly using polynomially many rational arithmetic operations and comparisons, with a bound polynomial in the original instance length and independent of the expanded encoding lengths of x and its Taylor coefficients. Then exact fixed-degree polynomial-observable comparisons at the minimizer of f over P belong to deterministic $\mathsf{P}^{\text{PosSLP}}$.*

Only addition, subtraction, multiplication, division by nonzero quantities, and comparisons are permitted in this interface. A polynomial bit-complexity QP algorithm alone does not meet it.

Let M be a polynomial-bit bound on Hessian Lipschitz variation on the bounding box and set $K = \max\{1, M/(2\mu)\}$. The first-order inequalities for $N(x)$ and p give

$$\|N(x) - p\| \leq \frac{M}{2\mu} \|x - p\|^2.$$

This is the established exact proximal Newton estimate [7, Theorem 3.4], specialized to the indicator of P . It requires neither strict complementarity nor a previously identified face. A feasible warm start of objective error at most $\min\{1, \mu/(8K^2)\}$ therefore gives

$$\|x_j - p\| \leq K^{-1} 2^{-2^j}, \quad x_{j+1} = N(x_j). \quad (23)$$

Retain every rational operation as a shared numerator-denominator circuit. The assumed QP algorithm then adds polynomially many gates per step; PosSLP answers its comparisons without expanding the fractions. The original rational KKT system, augmented by $z = h(x)$, has singleton real projection $\{h(p)\}$. One-block elimination gives a uniform alternative $h(p) = 0$ or $|h(p)| \geq 2^{-2^{a(L)}}$ for an effective polynomial a . Polynomially many steps in (23) make the observable error less than one quarter of that gap, after absorbing a polynomial-bit bound on $\|\nabla h\|$ on the box. Repeated squaring represents the gap, and final circuit comparisons identify the sign. QP decisions depend on oracle answers, so this is a Turing reduction, not the single-instance reduction of Section 4. The [transfer companion](#) provides the complete constants and circuit construction.

5.3 Boxes with structured Hessians

For a symmetric matrix H , let its comparison matrix be $B_{ii} = H_{ii}$ and $B_{ij} = -|H_{ij}|$ for $i \neq j$.

Corollary 5.3 (Structured boxes). *Let f satisfy the preceding global curvature promise and have fixed degree. Empty boxes are detected first. On a nonempty rational box P , suppose the comparison matrix of $\nabla^2 f(x)$ is positive definite for every $x \in P$. Exact fixed-degree observable comparisons and recovery of all active bounds are in $\mathsf{P}^{\text{PosSLP}}$. Sufficient cases are a supplied forest containing all Hessian interactions, or nonpositive off-diagonal Hessian entries throughout P . The dimension and number of active bounds are unrestricted.*

Pang and Han [9, Algorithm I and Proposition 2.1] solve the box QP in at most $2n$ pivots given $v > 0$ satisfying $H_J^{-1}v_J \geq 0$ for every principal subset J . Degeneracy is allowed. If $B \succ 0$, their comparison-matrix construction gives

$$d = B^{-1}\mathbf{1} > 0, \quad v = (H + B)d/2 \geq \mathbf{1}.$$

It uses rational linear systems and sign tests only. For a forest, changing coordinate signs along each tree makes B congruent to H , so $B \succ 0$. For a positive definite matrix with nonpositive off-diagonal entries, all principal inverses are nonnegative and $v = \mathbf{1}$ suffices. The transfer theorem now applies to circuit-valued Hessians and costs.

The structural promises are separate from their recognition. Missing forest interactions are checked by coefficient identities; no general polynomial-time recognition of the other promises is asserted. Dense graph objectives $\frac{\mu}{2} \sum_i x_i^2 + \sum_{ij} w_{ij}(x_i - x_j)^4 + c^\top x$, with $w_{ij} \geq 0$, supply an explicit family satisfying the second condition. Its box constraint rank may be n .

5.4 Separable polynomial flows

Corollary 5.4 (Network-flow polyhedra). *Let A be a directed node-arc incidence matrix and $P = \{x : Ax = b, \ell \leq x \leq u\}$ have rational data. Let $f(x) = \sum_e f_e(x_e)$, with each f_e an explicit polynomial of degree at most four and a supplied rational $\mu > 0$ satisfying $f_e'' \geq \mu$ on $\mathbb{R}$. After testing rational feasibility, exact value, coordinate, and fixed-degree observable comparisons belong to deterministic $\mathsf{P}^{\text{PosSLP}}$. Constraint rank is unrestricted; redundant balance rows, fixed arcs, and infinite capacity endpoints are allowed.*

For finite capacities, the Taylor subproblem is a separable quadratic flow with arc cost

$$\frac{1}{2}f_e''(x_e)y_e^2 + (f_e'(x_e) - f_e''(x_e)x_e)y_e.$$

Végh [12, Theorem 20] gives a strongly polynomial exact quadratic-flow algorithm using elementary rational arithmetic and comparisons, including capacitated instances. Its operation count is independent of coefficient encoding length, which supplies the transfer interface. Capacity gadgets and zero-cost auxiliary arcs are included in that source. The [source-interface review](#) also bounds artificial-arc costs by rational circuits, avoiding any use of expanded coefficient lengths. For quartics, the univariate curvature promise itself is checkable by rational quadratic nonnegativity tests.

Infinite endpoints in both corollaries reduce to finite ones. Find a polynomial-bit rational feasible point $\bar{x}$. Strong convexity and $f(p) \leq f(\bar{x})$ give

$$\|p - \bar{x}\| \leq 2 \|\nabla f(\bar{x})\|_1 / \mu.$$

Intersect the bounds with the coordinate box of this radius plus one around $\bar{x}$. It has polynomial encoding length, preserves the optimizer, and introduces no active artificial bound. The [flow companion](#) also proves the extension to every fixed polynomial degree. None of these results promises rational coordinates for the exact optimizer.

5.5 The remaining general boundary

The unresolved step in the Newton route is exact QP with an arbitrary circuit Hessian over an arbitrary rational polyhedron. Known QP arithmetic bounds that retain dependence on expanded Hessian encoding do not supply the transfer hypothesis. The hypothesis is sufficient; no equivalence with strongly polynomial QP is claimed.

A direct penalty argument does not bypass this issue. For $\min_{x \geq 0} (x+1)^2/2$, the squared-hinge penalty with coefficient $\lambda/2$ has minimizer $-1/(1+\lambda)$. Reaching error g requires $\lambda \geq g^{-1} - 1$. A short circuit for that coefficient does not make it an explicitly encoded polynomial-bit coefficient for a warm-start algorithm. Likewise, an iteration bound polynomial in $\log(1/\varepsilon)$ becomes exponential when $\varepsilon = 2^{-2^{a(L)}}$. These are failed interfaces, not hardness results. The [boundary note](#) records their precise scope.

The essential-variable construction, Newton recurrence, box QP, and flow QP are established ingredients. The additional statements here are their specified parameterized exact-sign and arithmetic-circuit consequences. The [primary-source comparison](#) and [targeted verification record](#) separate those claims from their dependencies and document independent internal review. They do not establish publication priority.

6 Full optimizer output for residual-convex cubic recourse

Consider an objective with a small group of core variables v and many residual variables z . Convex residual problems give efficient value queries at rational cores. They do not by themselves give a point near the residual optimizer at an unknown exact optimal core. This section closes that gap for cubics on product boxes, using one finite perturbation of the core coefficients. The result strengthens the earlier cubic completion theorem by removing its supplied joint quadratic core convexifier. The [complete companion proof](#) and its [independent review](#) record the inherited interfaces and their verification scope.

6.1 The output contract

Let $F \in \mathbb{Q}[v, z]$ have total degree at most three on $[0, 1]^k \times [0, 1]^n$. Assume that $F(v, \cdot)$ is convex for every v , and supply rational $L \geq 0$ with $F_{v_i v_i} \leq L$ throughout the box. Let $\sigma > 0$ be rational. The input length I includes these data and any structural certificates; polynomial certificate verification is assumed, or its actual cost must be added. Fixed coordinates may first be substituted and positive-width boxes normalized. Perturb only the core:

$$F_\gamma(v, z) = F(v, z) + \gamma^T v, \quad \gamma \in [-\sigma, \sigma]^k.$$

For each draw, let a be the lexicographically first globally optimal core. Define

$$S_a = \arg \min_{z \in [0, 1]^n} F(a, z), \quad p = \arg \min_{z \in S_a} \|z\|^2.$$

Compactness and residual convexity make p unique.

Theorem 6.1 (Full cubic recourse point oracle). *There is a base-computable endpoint-inclusive uniform product grid with M labels per coordinate and $\log M = \text{poly}(I)$ such that, for every draw and every integer $q \geq 0$, an evaluator returns a rational feasible (v_q, z_q) satisfying*

$$\|(v_q, z_q) - (a, p)\| \leq 2^{-q}.$$

Expected bit work and proof/output length are

$$f(k)(1 + L/\sigma)^k \text{poly}(I + q), \quad (24)$$

where f is computable and the polynomial exponent is absolute. One random work factor controls all precision queries. The noise law and selector are fixed before those queries. A certified gap for the sampled objective can be returned at the same order of cost. No joint convexifier, residual strong-convexity modulus, or unique residual optimizer is required.

For $n = 0$, this is the existing selected-core theorem. For $k = 0$, Theorem 3.1 supplies minimum-norm point output. Assume $k, n \geq 1$ below. We use three established interfaces: the [selected-core oracle](#), polynomial-time certified convex value optimization at rational cores, and the [exact polynomial fallback](#). The first returns short rational approximations to a with an all-precision work factor of the form (24). Its mesh requirements are lower bounds fixed from base data. The third has cost $B_0 \text{poly}(I + b + q)$, where $B_0 = 2^{\text{poly}(I)}$ is independent of sampled coefficient length b and requested precision q .

6.2 A sound certificate for exact core endpoints

Put $\bar{L} = \max\{1, L\}$ and fix a base rational $0 < t \leq 1/2$. For each i , query selected cores of the two auxiliary objectives $F_{\gamma - te_i}$ and $F_{\gamma + te_i}$ to coordinate error $r \leq t/(8\bar{L})$. Let $[\ell_i^-, u_i^-]$ and $[\ell_i^+, u_i^+]$ be the resulting certified intervals. Use

$$u_i^- < t/\bar{L} \implies v_i = 0, \quad \ell_i^+ > 1 - t/\bar{L} \implies v_i = 1. \quad (25)$$

Each conclusion holds for every original global optimizer, including on tied draws.

To prove this, hold γ_{-i} fixed and project out all other variables:

$$W(s) = \min_{v_{-i}, z} \{F(v, z) + \gamma_{-i}^T v_{-i} : v_i = s\}.$$

The function $W(s) - \bar{L}s^2/2$ is concave, since each function in the minimum has this property. If a_i minimizes $W(s) + \gamma_i s$ and b_i minimizes $W(s) + (\gamma_i - t)s$, their optimality inequalities imply $a_i \leq b_i$. At an interior minimizing contact, semiconcavity gives differentiability, with derivative equal to minus the tilt. Thus, if both coordinates are interior,

$$t = W'(b_i) - W'(a_i) \leq \bar{L}(b_i - a_i).$$

The first strict interval test contradicts this inequality whenever $a_i > 0$. Reflection proves the other test.

Failure to recognize an actual endpoint has a short conditional noise interval. Set

$$\theta = \sup_{s > 0} \frac{W(0) - W(s)}{s}.$$

Coefficient bounds make W Lipschitz, so θ is finite. An original zero coordinate requires $\gamma_i \geq \theta$. If $\gamma_i > \theta + t$, every negatively shifted optimizer has coordinate zero, and the first test passes. A missed lower endpoint therefore requires $\gamma_i \in [\theta, \theta + t]$; the upper endpoint has the same bound. On an endpoint-inclusive M -label grid, an interval of length t has mass at most $t/(2\sigma) + 2/M$. Consequently

$$\Pr\{\text{some endpoint of the selected core is missed}\} \leq kt/\sigma + 4k/M. \quad (26)$$

Unforced coordinates are provisionally free. A later certificate rejects any that are actually endpoints; probability is not used to justify a face declaration.

6.3 Cubic fibers and a lattice certificate for their minors

Let A be the certified face, with m free coordinates and relative center c_A , and let $c_z = (1/2, \dots, 1/2)$. Define

$$H_A = \nabla_{zz}^2 F(c_A, c_z), \quad g_A(v) = \nabla_z F(v, c_z), \quad B_A(v) = \begin{bmatrix} H_A \\ g_A(v)^T \\ I_n \end{bmatrix}.$$

Write $H(v, z) = \nabla_{zz}^2 F(v, z)$. This matrix is affine and positive semidefinite. Reflection about (c_A, c_z) gives $\ker H_A \subseteq \ker H(v, z)$ on the whole face product, and $\ker H(v, c_z) = \ker H_A$ for $v \in \text{relint } A$. Individual residual boundary points can have larger kernels. Provided $a \in \text{relint } A$, as acceptance below certifies, the cubic argument behind Theorem 3.1 gives, for any $y \in S_a$,

$$S_a = \{z \in [0, 1]^n : H_A(z - y) = 0, g_A(a)^T(z - y) = 0\}. \quad (27)$$

Only the row $g_A(v)^T$ varies with v , and it has degree at most two. Every square minor of $B_A(v)$ therefore has degree at most two.

Compute integers $D, B \geq 1$, with polynomial bit length, such that every minor on every original face, multiplied by D , is an integer polynomial of coefficient height at most B . This requires only denominator and coefficient bounds. For example, let d clear all possible entry coefficients, and let C bound their cleared heights. With $s_* = (k+1)(k+2)/2$, the conservative choices

$$D = d^m, \quad B = D n! (s_* C)^n$$

suffice. Neither minors nor faces are enumerated.

For the free coordinates of a , let $\phi(a)$ list all monomials of degree at most two, including one; its length is $s = (m+1)(m+2)/2$. Choose an integer scale T and obtain a short dyadic b with $\|b - a\|_\infty \leq 1/(4T)$. Put $w_j = \text{round}(T\phi_j(b))$. Since these monomials are 2-Lipschitz on the cube, $|w_j/T - \phi_j(a)| \leq 1/T$. Apply LLL reduction with parameter $3/4$ [8] to the rank- s integer lattice generated by $[I_s \mid w]$. If ℓ_1 is its first reduced vector, accept only when

$$\|\ell_1\|^2 > 2^{s-1} (4sB)^2. \quad (28)$$

The usual inequality $\|\ell_1\| \leq 2^{(s-1)/2} \lambda_1$ then proves

$$|h^T \phi(a)| > sB/T \quad (0 < \|h\|_\infty \leq B, h \in \mathbb{Z}^s). \quad (29)$$

Indeed a violating h would give the nonzero lattice vector $(h, h^T w)$ of length at most $\sqrt{5} sB < 4sB$. LLL and its reduced-basis certificate have cost polynomial in $s + \log T + \log B$; there is no enumeration of the tested polynomial family.

For the success analysis, enlarge the height to $R_s = 2^s 4sB$. If every nonzero polynomial of height at most R_s has value at least η in absolute value, then the test accepts whenever

$$T\eta > (s+1)R_s. \quad (30)$$

A lattice vector with $\|h\| > R_s$ is already long. Otherwise its last coordinate has magnitude at least $T\eta - sR_s > R_s$. This proves the sufficient condition. Enlarging the family is necessary because a short vector can have height greater than B .

The coordinate polynomials v_i and $1 - v_i$ belong to the tested family. Acceptance thus certifies that all free coordinates are truly interior. With $\xi = sB/T$, set

$$\delta = \min\{1/2, \xi\}, \quad \mu = \min\{1, \xi/D\}.$$

Every free coordinate is at least δ from its endpoints, and every nonzero minor at a has magnitude at least μ . Identically zero minor polynomials cause no difficulty. For a vertex face, use $\delta = 1/2$ and $\mu = 1/D$ without LLL.

The [conditional cubic fiber bound](#) now computes a rational $\Gamma \geq 1$, of bit length polynomial in $I + \log T$, such that

$$\text{dist}(z, S_a) \leq \Gamma[F(a, z) - \min_w F(a, w)]^{1/4} \quad \text{when the gap is in } [0, 1]. \quad (31)$$

Its proof combines the cubic transverse estimate with a Hoffman bound for (27). The supplied nonzero-minor margin replaces the integer-minor bound for rational coefficients; the face margin bounds the positive eigenvalues transverse to $\ker H_A$. Rank-zero fibers are included. The lattice step supplies these premises without knowing exact algebraic coordinates of a .

6.4 Smoothed margins and the finite-grid transfer

Choose a rational $U \geq 1$ with $\nabla_{vv}^2 F \preceq UI$ on the box. Its numerical size will enter only the base precision, through $\log U$. On any core face, the projected value $V(v) = \min_z F(v, z)$ satisfies that $V(v) - U \|v\|^2 / 2$ is concave. Let C be the interior points having an affine lower support on the whole face. At $a \in C$ the support slope g_a is unique and

$$0 \leq V(x) - V(a) - g_a^T(x - a) \leq (U/2) \|x - a\|^2.$$

For $a, b \in C$, put $r = \|a - b\|$. Comparing the upper support at a and lower support at b , then choosing $h = r(g_b - g_a) / \|g_b - g_a\|$ when this displacement stays in the face, gives

$$(g_b - g_a)^T h \leq (U/2)(r^2 + \|h\|^2), \quad \|g_b - g_a\| \leq U \|a - b\|.$$

Thus the contact-gradient map is locally U -Lipschitz. A countable interior cover and a disjoint measurable partition give $|g(E \cap C)| \leq U^m |E|$. An interior optimizer under tilt has this contact slope, with the opposite sign. Continuous uniform noise therefore puts probability at most $(U/(2\sigma))^m |E|$ on such cores. No continuity of a residual optimizer is asserted.

For every nonzero integer polynomial p of degree at most two,

$$|\{v \in [0, 1]^m : |p(v)| \leq \eta\}| \leq 8\sqrt{\eta}, \quad 0 < \eta < 1. \quad (32)$$

A nonzero square coefficient permits coordinate slicing, using the univariate quadratic sublevel length bound $4\sqrt{\eta/|c|}$. If only mixed quadratic terms occur, slice along $(e_i + e_j)/\sqrt{2}$ for a nonzero mixed coefficient. The leading magnitude is at least $1/2$ and the projected cube volume is $\sqrt{2}$, giving the stated constant. Linear and constant polynomials are easier.

Put $R = 2^{s_*} 4s_* B$ and

$$A_* = 8 \cdot 3^k \max\{1, U/(2\sigma)\}^k (2R + 1)^{s_*}.$$

A union bound over faces and all degree-two integer polynomials of height at most R bounds the continuous bad-margin probability by $A_* \sqrt{\eta}$. The logarithm of this family size is polynomial; the algorithm never lists its members.

For a fixed face and polynomial, the bad event says that there exists an interior face optimizer with $|p(a)| \leq \eta$. It is expressed by one existential full-variable witness block and one universal competitor block, with degree at most three. The inherited [finite-noise section bound](#) therefore gives at most $2^{\text{poly}(I)}$ interval or point components in every one-dimensional noise section, uniformly in the fixed values of the other coefficients. Replacing continuous marginals by grid marginals one

at a time changes each section probability by at most twice its component count divided by M . Summing over the family gives

$$\Pr_{\text{grid}}\{\text{bad margin}\} \leq A_*\sqrt{\eta} + C_*/M, \quad C_* = 2^{\text{poly}(I)}. \quad (33)$$

Exact-zero atoms are included. This argument bounds a sufficient condition for LLL success; it does not need a semialgebraic description of the LLL algorithm. The [margin companion proof](#) provides the complete section-count construction.

6.5 One base certificate, one law, and a fixed-selector fallback

Choose the exact full-point fallback factor $B_0 \geq 2$ first, then set

$$t = \min\{1/2, \sigma/(16kB_0)\}, \quad \eta = \min\{1/4, (16A_*B_0)^{-2}\}.$$

Choose a power of two $T > (s_* + 1)R/\eta$. All these quantities have polynomial bit length and precede the noise mesh. Instantiate the selected-core interface for F and the $2k$ shifted baselines $F \pm tv_i$. Their input lengths remain polynomial in I , and their core second derivatives are unchanged. Use the same random vector γ for every instance. Choose one power-of-two M satisfying all these interfaces' lower mesh requirements and

$$(4k + C_*)/M \leq 1/(8B_0).$$

Hence $\log M = \text{poly}(I)$, independently of all later accuracy queries. The shifted queries are auxiliary problems on the same draw, not new random samples.

Run the face tests and one lattice test at the chosen scale. If the lattice test fails, use exact fallback on the original draw. Equations (26), (30), and (33) bound this failure probability by $1/(4B_0)$. A missed endpoint forces rejection because its coordinate polynomial vanishes. Every accepted certificate is valid independently of the probability estimate.

The exact fallback selects the same (a, p) . A fixed-block formula says that its witness is globally optimal and every equal-value competitor has lexicographically larger core, or the same core and residual squared norm at least that of the witness. Residual convexity makes the witness unique. The degree remains at most three, and the lexicographic comparison has linear-size Boolean syntax. The inherited singleton-coordinate construction thus costs $B_0 \text{poly}(I + b + q)$, with B_0 independent of b, q . Clipping the rational approximation to the box preserves its distance guarantee. Ties neither change the selector nor require resampling.

6.6 Completing the residual point at every precision

On acceptance, Γ in (31) has polynomial base bit length. Given $\varepsilon = 2^{-q}$, put $e = \varepsilon/2$ and choose rational $R_z \geq \max\{1, \sqrt{n}\}$ and $G \geq \max\{1, \sup \|\nabla_v F\|\}$. Set

$$\tau = \frac{e^6}{1024R_z^4\Gamma^4}, \quad \eta_q = \frac{\tau e^2}{8}, \quad d_q = \min\left\{\frac{\varepsilon}{2}, \frac{\tau e^2}{32G}\right\}.$$

Obtain a short rational $b \in A$ with $\|b - a\| \leq d_q$, fixing certified endpoints exactly, and solve the rational convex cubic

$$Q_b(z) = F(b, z) + \tau \|z\|^2$$

to objective gap at most η_q .

For the proof, let z_τ minimize $F(a, z) + \tau \|z\|^2$. Comparison with p gives $\|z_\tau\| \leq \|p\| \leq R_z$. If s is its nearest point in S_a and $d = \|s - z_\tau\|$, then the error bound and the minimum-norm projection inequality imply

$$\frac{d^4}{\Gamma^4} \leq F(a, z_\tau) - F(a, p) \leq 2R_z \tau d, \quad \|z_\tau - p\|^2 \leq 2R_z d.$$

The gap is initially at most $\tau R_z^2 \leq 1$, so the error bound applies. The chosen τ gives $d \leq e^2/(8R_z)$ and $\|z_\tau - p\| \leq e/2$. Uniformly in z , $|F(b, z) - F(a, z)| \leq Gd_q$. The returned point therefore has true regularized gap at most $\eta_q + 2Gd_q \leq \tau e^2/4$. Strong convexity places it within $e/2$ of z_τ , and hence within e of p . Together with the core error this proves the theorem's Euclidean bound. A finer point query and the inherited value lower bound provide the optional sampled objective-gap certificate.

All requested bit lengths are $\text{poly}(I) + O(q)$. Add the common random work factors of the original and $2k$ auxiliary core interfaces, and add B_0 times the indicator of the single base certificate-failure event. This sum controls every precision query and has expectation (24). Independence between work factors and certificate events is unnecessary. Large U , small certified margins, and Γ affect polynomial bit lengths rather than replacing L by a larger numerical parameter in the expected-work factor.

The product box and total degree three are essential to this argument. It covers, for example, $F(v, z) = vz^2$ and $F(v, z) = v^2(2 - z)$, which admit no finite quadratic core convexifier. For the latter, the minimum-norm residual selection jumps from zero at $v = 0$ to one at every positive core. The proof avoids assuming that selection is continuous. It does not extend the conclusion to arbitrary coupled domains, quartics, expanded algebraic output, or the unperturbed objective at the same cost.

7 Certificates and exact representations

An exact witness is useful only with a specified representation and checking contract. Section 4 separates expanded rational coordinates from shared arithmetic circuits. The same distinction applies to positive definite Gram matrices: an exponential lower bound on expanded denominator lengths can coexist with a polynomial-size shared circuit.

7.1 A compact interior Gram without a sign oracle

Theorem 7.1 (Circuit representation of an interior Gram). *Let $f \in \mathbb{Q}[X_1, \dots, X_n]$, $n \geq 1$, have degree at most four. Suppose its input includes a full rational positive definite matrix A with*

$$v^\top \nabla^2 f(X) v = (v, X \otimes v)^\top A (v, X \otimes v).$$

Let $z_2(X)$ list every monomial of degree at most two. A deterministic polynomial-time algorithm constructs a symmetric shared rational-circuit matrix Q such that

$$f(X) = z_2(X)^\top Q z_2(X), \quad Q \succ 0 \iff \min_{\mathbb{R}^n} f > 0.$$

The identity and all divisions in the construction are valid on every valid input, independently of the sign of the minimum. Construction requires neither a sign oracle nor a supplied positive objective margin.

Proof. Put $m = n + n^2$ and $\mu = \det(A)/(\text{tr } A)^{m-1}$, so $A \succeq \mu I$. The explicit observable

$$h(X) = f(X) - \frac{\|\nabla f(X)\|^2}{2\mu}$$

has degree at most six and polynomial encoding length. If p is the unique minimizer, then $h(p) = f(p)$. The constructive upper reduction of Theorem 4.1, stopped before its final sign query, gives a rational-circuit vector a for which

$$|h(a) - h(p)| \leq g/8, \quad h(p) \neq 0 \implies |h(p)| \geq g.$$

Thus $c = h(a) > 0$ whenever $f(p) > 0$. No sign of c is tested.

Set $d = X - a$, $b = \nabla f(a)$, and $\bar{A} = A - \mu \text{diag}(I_n, 0)$. Define coefficient matrices by

$$(d, a \otimes d) = C_U z_2, \quad (0, d \otimes d) = C_V z_2, \quad d + b/\mu = C_{\text{aff}} z_2.$$

Their dimensions and their shared circuit sizes are polynomial in the original input length. If e_0 selects the constant monomial, put

$$Q = \frac{1}{2}(C_U + C_V/3)^\top \bar{A}(C_U + C_V/3) + \frac{1}{36}C_V^\top \bar{A}C_V \\ + \frac{\mu}{2}C_{\text{aff}}^\top C_{\text{aff}} + ce_0e_0^\top.$$

Taylor's integral identity and completion of the gradient square give $f = z_2^\top Q z_2$ exactly. In particular, a denotes an exact rational vector even though it approximates an optimizer.

Since $\bar{A} \succeq \mu \text{diag}(0, I_{n^2})$, the first three summands are positive semidefinite. When $c > 0$, the second summand controls all translated homogeneous quadratics $d_i d_j$, the third controls the affine factors $d_i + b_i/\mu$, and the last controls the constant. Those polynomials span the full space of degree at most two. Their Gram sum is therefore positive definite. Conversely, $Q \succ 0$ would imply $f(X) \geq \lambda_{\min}(Q) \|z_2(X)\|^2 \geq \lambda_{\min}(Q) > 0$. This proves the equivalence. Every matrix operation appends polynomially many gates while retaining the shared circuit for a . $\square$

The [complete companion argument](#) and its [independent review](#) give the exact Taylor constants and representation contract. The expanded positive definite Gram lower bound in Theorem 4.5 still applies. The entries of the circuit matrix can still require exponentially many bits when expanded.

It also does not give ordinary polynomial-time verification of circuit-encoded identities or positive definiteness. In fact the equivalence links positive definiteness of this constructed circuit matrix to exact optimization. Nor does it produce a polynomial-size unweighted rational SOS factorization: decomposing circuit-encoded positive weights into rational squares is a separate task.

7.2 A reference checker for the three approximation contracts

The [reference implementation](#) makes the output distinctions executable with exact rational arithmetic. It accepts a sparse rational polynomial, a rational box, optional additional rational linear inequalities, and an explicit identity

$$f(x) = a^\top x + b + \sum_j w_j (u_j^\top x + v_j)^{2d_j}, \quad w_j \geq 0, \quad d_j \geq 1.$$

Coefficient comparison verifies the identity and hence global convexity. This is a sufficient certificate language; general convexity recognition is not included. Expanding a succinct factored identity can be expensive, so no polynomial complexity claim in the factored encoding is made.

A witness provides a feasible candidate and a tangent anchor. The checker forms a supporting affine function and minimizes it over the box to obtain a justified lower bound on the constrained optimum. It accepts no asserted optimal value. Its three entry points then check, respectively:

1. an objective-gap certificate;
2. distance to the optimizer set using an error-bound constant derived from the checked polynomial and domain;
3. distance to the fixed minimum-norm optimizer, using a checked regularized objective and the effective selector schedule.

The gradient coefficient matrix is extracted exactly; its constant row is retained so that an affine slope is not mistaken for an invariant direction. Every comparison uses rational arithmetic. Floating-point inputs are rejected.

These checkers are sound but incomplete even in their supported class. A box tangent may be too weak for an optimum determined by an extra linear inequality. Rejection says that the supplied certificate is insufficient; it does not prove that the candidate is inaccurate. A regression test records this case. The implementation searches for no optimizer or witness, does not implement the unbounded-domain reduction, and does not solve PosSLP. Its role is to make the contract distinctions testable without presenting a small example as a general optimization solver.

The ten focused tests cover nonunique minimizers, a tiny value gap with large coordinate error, fixed minimum-norm selection, lower-dimensional domains, affine objectives, incorrect convexity identities, and infeasible points. The deterministic demonstration also shows thirteen shared recurrence nodes whose expanded rational denominator has 4,097 bits. These finite checks support the implementation and the displayed identities; the proofs establish the asymptotic theorems.

8 What the results settle and what remains open

The results support one integrated account because their proofs share the same arithmetic distinction. A value algorithm becomes a point algorithm only after an effective distance modulus is supplied. A point algorithm becomes an exact comparison algorithm only after a separation bound is combined with a representation that can reach the required precision. An exact circuit can be short even when printing its value is expensive.

Completed extensions. The global point theorem treats arbitrary rational polyhedra, sparse input, and numerical degree dependence, including finite attainment and a fixed original-coordinate minimum-norm selector. The residual-convex cubic theorem removes the joint core convexifier on product boxes and supplies one finite noise law for every precision. The constrained exact results remove the constraint-rank restriction under an alternative nonlinear-dimension parameter or under verified box/flow structure. The unconstrained exact upper theorem extends to explicit unary-degree polynomials and observables. The circuit-Gram theorem supplies a compact exact representation that complements the expanded-output lower bounds.

The earlier convex cubic point theorem, quartic point reductions, exact quartic sign completeness, rational-optimizer hardness, irrational zero example, and witness-size bounds were audited and integrated. They are historical results in this repository, not results first created by this completion pass. The accompanying [result map](#) links each theorem, its proof, its independent review, and its verification record.

The unrestricted constrained comparison question. This report does not establish deterministic P^{PosSLP} for an arbitrary globally strongly convex quartic over an arbitrary rational polyhedron. The constrained Newton recurrence itself is available. What the present argument lacks is an exact

quadratic-subproblem algorithm whose number of arithmetic operations is polynomial independently of the expanded bit lengths of circuit-encoded Hessian entries. Ordinary polynomial-bit QP algorithms do not supply that interface. The new structural results verify it in concrete families; the nonlinear-dimension theorem uses a different route. The [boundary analysis](#) also explains why straightforward smooth penalties and accuracy-driven barriers do not resolve the unrestricted case. This is a remaining question, not an impossibility result for every possible method.

Other limits. The cubic recourse proof uses the product-box geometry and the quadratic degree of its relevant minor polynomials. It does not settle arbitrary coupled feasible sets or quartic recourse. The global point theorem requires global convexity and polynomial dependence on numerical degree; it does not give a polynomial-input-length result for exponentially large binary-encoded degrees or arithmetic-circuit polynomial input. Equality in the exact quartic class has the stated upper reduction, without a matching hardness claim here. At a zero minimum, rational SOS and real SOS remain different questions. Compact Gram construction does not imply inexpensive exact validation or a short expanded certificate.

Prior work and source limits. The source comparison credits prior value optimization, effective radii, invariant directions, error-bound theory, essential-variable extraction, Newton convergence, and structured QP algorithms. Different output wording alone is not evidence of a new theorem. In particular, the degree-based error exponent is not claimed as new. The full primary text of Yang’s 2009 paper could not be obtained in this audit, and the inspected version of *Hesse’s Redemption* is arXiv version 1, not its proceedings version. The [versioned source ledger](#) records those limits. Source-level proof-transfer cautions are subordinate to the theorem comparison and do not assert that the source’s main results fail. Publication priority remains unestablished.

Verification and completion. Each new proof has an internal independent review, and the existing central results have fresh scoped audits. The saved exact-arithmetic diagnostics exercise identities, degeneracies, and certificate boundaries. The report and reference implementation have their own targeted checks. No finite test establishes a universal complexity bound, and internal research-agent review is not external peer review. The [verification record](#) lists the commands actually run and their scope; project-wide checks and CI were not duplicated.

This completes the documented development and review program. It does not make the broader subject mathematically exhausted: the unresolved cases above remain explicit research questions, and they are excluded from every completed theorem statement.

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